Start here

Introduction

Complete beginner course for AI development

Welcome to Python for AI

Learn Python from scratch and start building real AI applications. This is the fastest, beginner-friendly course for Python and AI development.

Whether you've never written a line of code or you're switching from another language, this course takes you from zero to confidently building Python projects for AI. No prior experience needed.

This handbook accompanies the **Complete Python for AI Course**, giving you everything in written form with extra details.

Complete beginner

No programming experience required

AI focused

Python specifically for AI development

Hands-on learning

Practice with real code examples

What you'll learn

Master the Python fundamentals essential for AI development:

Set up like a pro

Master VS Code, Git, and the exact workflow used in real AI companies

Learn Python fast

Variables, functions, APIs, and everything you need for AI development

Build AI agents

Apply what you learn to create real AI-powered applications

Welcome to Python for AI

I'm **Teja Sai Mahesh Ch**, and I'll be your guide on this journey.

I started Working with Python in 2019 while I am working with "Waymo" as a Associate.

- System Engineer
- AWS Dev
- AI Engineer
- Lead AI Engineer
- AI Architect
- AI Product Manager

https://www.linkedin.com/in/teja-sai-ai-pm/

https://x.com/TChennamsetti

Is this for you?
Find out if this course matches your goals

Who this is for

This course works whether you've never written a line of code or you're coming from another programming language.

This course is for you if:

- You want to build AI applications and need Python skills to make that happen.
- You prefer learning by doing rather than spending hours on theory.
- You're ready to install software, write code, and build real projects.

About the code examples

As we progress, We will use AI Vibe Coding tools to write code for us. The goal isn't to understand every single line immediately - it's exposure to what's possible. We start simple and build complexity over time.

Use AI to learn faster

ChatGPT, Notebook Lm(Finetune) Or create your own AI assistants as your tutors

Example prompts:

"Explain this Python code in simple terms: [paste code]"

"What does this error mean? [paste error message]"

"Show me 3 different ways to solve this problem in Python"

Watch actively and code along. Make mistakes and fix them. Use the handbook to go deeper and copy code. Ask AI when stuck. After each major section, try building something similar on your own. Accept that you don't need to master everything before moving forward.

Before you begin

Why Python

The perfect language for AI development

The best choice for AI

Python is the dominant language in AI research and development: OpenAI, Google, Anthropic (and many others) use Python heavily for prototyping, experimentation, and tooling around their LLMs.

It has also been the backbone of modern machine learning, powering the entire ecosystem of frameworks like TensorFlow, PyTorch, scikit-learn, and XGBoost – the tools that made deep learning breakthroughs possible.

Python dominates AI

Easiest to learn

Python reads almost like English

AI libraries

TensorFlow, PyTorch, scikit-learn built-in

Industry standard

Used by every major AI company

Huge community

Millions of developers and resources

Simple yet powerful

Compare Python to other languages:

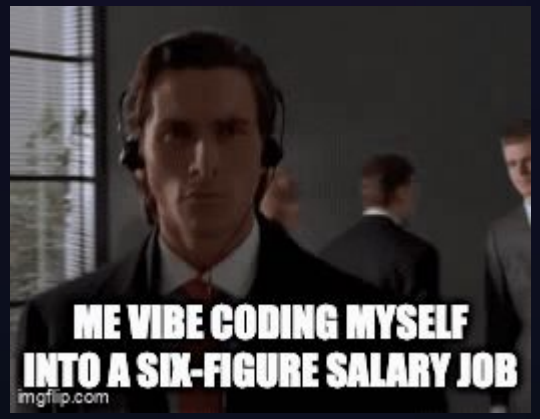
```python
# Python: Calculate average
numbers = [10, 20, 30, 40, 50]
average = sum(numbers) / len(numbers)
print(f"Average: {average}")
```

```java
// Java: Same task, more complexity
public class Average {
    public static void main(String[] args) {
        int[] numbers = {10, 20, 30, 40, 50};
        int sum = 0;
        for (int num : numbers) {
            sum += num;
        }
        double average = (double) sum / numbers.length;
        System.out.println("Average: " + average);
    }
}
```

The best language to learn

AI has created many opportunities in the job market:

- **High demand:** Python AI developers earn $130,000+ starting salaries
- **Remote work:** Code from anywhere in the world
- **Growing field:** 40% annual growth in AI jobs
- **Creative freedom:** Build things impossible before



Essential for AI work

Every AI application uses Python:

- Language models (ChatGPT, Claude) {Built with Python using transformers and neural networks}
- Computer vision (Tesla Autopilot) {Image recognition and processing with OpenCV and PyTorch}
- Data analysis (Netflix recommendations) {Process millions of data points with pandas and NumPy}
- Machine learning (Spotify playlists) {Create intelligent systems with scikit-learn and TensorFlow}

Using AI assistants

Learn how AI tools accelerate your Python learning

Before you begin

AI assistants

Use AI tools to learn Python faster

Learning with AI assistants

AI tools can accelerate your Python learning when used correctly. Here's how to use them effectively.

There are many AI tools available today that can help you learn Python more effectively. Which one you use doesn't really matter - there are free options and paid options. Some are simple chat interfaces, others integrate directly into your code editor.

Pick a tool you already have or are familiar with. Explore paid options if you have the budget. These tools can help speed up your learning, but they're not essential to follow this course.

Best tools for learning

ChatGPT

General coding help and explanations

Claude

Complex problem solving and debugging

GitHub Copilot

Code completion in your editor

Codex, Antigravity, Ai studio, Lovable, Bolt, Cursor, Claude Code .....

AI-powered code editors : will be covered in upcoming modules

How to ask for help

Bad prompt:

My code is not working, plz fix

Good prompt:

I'm learning Python and trying to create a list of numbers from 1 to 10.

My code: numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

Is there a more efficient way?

Effective strategies

1. **Ask for explanations**
   “Explain Python functions in simple terms”

2. **Debug with context**
   Share your code and the exact error message

3. **Request alternatives**
   “Show me 3 ways to solve this problem”

4. **Verify understanding**
   Test AI suggestions and ask follow-up questions

AI assistants are tools to enhance learning, not replace it. The goal is understanding, not just working code.

Create your personal Python tutor

Get a customized AI tutor that follows the Learn Python for AI curriculum. Use this system prompt:

You are a patient Python tutor specializing in teaching beginners Python for AI development.

Follow these guidelines:

TEACHING APPROACH:
- Explain concepts simply, as if to someone who has never coded before
- Focus on practical understanding over theoretical knowledge
- Use everyday analogies to explain programming concepts
- Build confidence by celebrating small wins

RESPONSE STRUCTURE:
1. First, acknowledge what the student is trying to do
2. Explain any errors in simple terms
3. Provide a corrected solution with clear comments
4. Explain why this approach works
5. Suggest one next step to build on their learning

CODE STYLE:
- Write clean, simple Python code
- Add comments explaining what each line does
- Use descriptive variable names
- Keep examples under 10 lines when possible

IMPORTANT RULES:
- Never provide solutions without explanations
- If asked to "just fix it," still explain what was wrong and why
- Encourage students to understand errors as learning opportunities
- When students are stuck, guide them with questions rather than immediate answers

Remember: The goal is understanding, not just working code. Every interaction should build the student's confidence and skills.

Best tools for learning

ChatGPT

Requirements: ChatGPT Plus ($20/month)

1. Open Chat GPT and click Explore GPTs
2. Click Create
3. In the Configure tab, paste the prompt above in the Instructions field
4. Name it “Python Tutor”
5. Save visibility as Only me

Claude

Requirements: Claude Pro ($20/month)

1. Click Projects in the sidebar
2. Click Create project
3. Name it “Python Tutor”
4. Click Create project to start
5. Paste the prompt in Set project instructions

Free Versions

No subscription required

1. Open Chat GPT or Claude
2. Start a new conversation
3. Paste the entire prompt above
4. Add your Python question after the prompt
5. The AI will follow the tutor guidelines

Note: Please Use Chat GPT Go Free or Gemini Free with JIO or Perplexity

Python setup

Getting started

Set up your Python development environment

Here's what I'll cover

In this section, you'll set up everything needed to start writing Python code. We'll install Python, configure your code editor, and run your first program.

What is Python

Understanding programming basics

Install Python

Get Python on your computer

First program

Write and run your first code

3. **Computer executes** those instructions

Python setup

What is Python

Understanding programming and Python basics

What is programming?

Programming is writing instructions for computers to follow. Just like you might write a recipe for someone to cook a meal, you write code to tell a computer what to do.

Computers are powerful but not smart. They need extremely detailed, step-by-step instructions for everything. A programming language is how we write these instructions in a way both humans and computers can understand.

Why do we need programming languages?

Computers only understand binary (1s and 0s). Writing in binary would be impossible for humans, so we use programming languages as a middle ground:

1. **You write** code in Python (looks like English)
2. **Python translates** your code into instructions

What makes Python special?

Readable code

Python looks almost like plain English

Beginner friendly

Designed to be easy to learn

Versatile

Works for websites, AI, automation, and more

Huge community

Millions of people ready to help

Your first taste of Python

```python
# This is Python code - it almost reads like English!
name = "Sarah"
age = 25
print(f"Hello, my name is {name} and I am {age} years old")

# Making a simple decision
if age >= 18:
    print("I can vote!")
else:
    print("I'm too young to vote")
```

Python versions explained

Python 3 (current version): This is what you'll learn and use.

Python 2 (outdated): Stopped being supported in 2020. Ignore it - it's like using Windows XP.

Always use Python 3 until the new version releases

Python setup

Installing Python

Get Python running on your computer

Choose your operating system

Installing Python differs based on your operating system. Select your platform below. Don't worry - Python works the same way once installed.

Windows

Windows 10 or 11

macOS

Mac computers

Linux

Ubuntu, Debian, etc.

Platform guides

Install on Windows

Installing Python on Windows 10/11

Download Python

1. Go to python.org/downloads
2. The site will detect you're on Windows and show the latest version
3. Click the download button to get the installer
4. Save it to your Downloads folder

Always get Python from the official website to ensure it's secure and up-to-date.

Install Python

This is the most important part - pay close attention to step 2!

1. Find and run the downloaded installer
2. **Important:** Check ✓ **"Add python.exe to PATH"** at the bottom

⚠️ If you miss the "Add to PATH" checkbox, Python won't work from Terminal. This is the #1 installation mistake.

1. Click "Install Now" (not "Customize installation")
2. If Windows asks for permission, click "Yes"
3. Wait for installation (usually 1-2 minutes)
4. Click "Close" when you see "Setup was successful"

Verify installation

Let's make sure Python installed correctly:

1. **Open Terminal:**
   - Press `Windows + R`
   - Type `wt` and press `Enter`
   - Or search "cmd" in Start menu

> Windows Terminal is the modern app recommended for Python on Windows.

2. Type this command:

```
python --version
```

3. You should see the Python version you just installed

> On Windows, you can usually use **python** (not **python3** like on Mac/Linux).

• Type exit() and press Enter

Test Python

Let's run Python for the first time:

1. In cmd, type:

```
python
```

2. You'll see something like:

```
Python 3.13.5 (tags/v3.13.5, Jul 29 2025) [MSC v.1935 64 bit]
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

3. The `>>>` means Python is running! Try:

```
print("Hello from Python!")
```

4. Press `Enter` to see:

```
Hello from Python!
```

5. To exit Python:

Type `exit()` and press `Enter`

Or Press `Ctrl + Z` then enter `Enter`

Troubleshooting

'python' is not recognized

This means Python wasn't added to PATH. Here's how to fix it.

Quick Fix: Try py instead of python:

```
py --version
```

1. Find and run the downloaded installer
2. **Important:** Check ✓ **"Add python.exe to PATH"** at the bottom

Solution 1: Reinstall Python
1. Uninstall Python from Settings > Apps
2. Download and run installer again
3. CHECK the "Add to PATH" box this time!

Solution 2: Add to PATH manually
1. Search "Environment Variables" in Start menu
2. Click "Environment Variables" button
3. Under "System variables", find and double-click "Path"
4. Click "New" and add these two paths:

```
C:\Users\[YourName]\AppData\Local\Programs\Python\Python313
C:\Users\[YourName]\AppData\Local\Programs\Python\Python313\Scripts
```

5. Replace [YourName] with your Windows username
6. Replace Python313 with your actual Python version folder
7. Click OK on all windows
8. Close and reopen Terminal

Permission denied,

During installation:

- Right-click the installer
- Select "Run as administrator"

When running Python:

- Open Terminal as administrator
- Right-click Terminal > Run as administrator

Windows Defender blocking Python

Windows might flag Python as suspicious:

1. When warning appears, click "More info"
2. Click "Run anyway"
3. Or add Python to Windows Defender exceptions:
   Settings > Windows Security > Virus & threat protection
   Add an exclusion > Folder
   Add Python installation folder

Multiple Python versions

Windows can have multiple Python versions. To manage them:

See all versions:

```
py -0
```

See all versions:

```
py -3.13 --version
py -3.12 script.py
```

Set default version: Create py.ini in your home folder with:

```
[defaults]
python=3.13
```

Microsoft Store Python

If typing python opens Microsoft Store:

1. Settings > Apps > Apps & features
2. Click “App execution aliases”
3. Turn OFF the Python aliases
4. Use the python.org installer instead

The Microsoft Store version has limitations for development.

Multiple Python versions

Windows can have multiple Python versions. To manage them:

See all versions:

```
py -0
```

See all versions:

```
py -3.13 --version
py -3.12 script.py
```

Set default version: Create py.ini in your home folder with:

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[defaults]
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Platform guides

Install on macOS

Installing Python on Mac computers

Check existing Python

macOS includes an older Python version for system use. Let's check what you have:

1. **Open Terminal:** `Cmd + Space`
   - Press to open Spotlight
   - Type "Terminal" and press Enter
   - Or find Terminal in Applications > Utilities
2. **Check Python version:**

```
python3 --version
```

If you see a recent Python 3 version, you might already be set!

Some Mac users can use just `python` instead of `python3`, but this depends on your system configuration. When in doubt, use .

Download Python

1. Go to **python.org/downloads**

2. The site will detect you're on macOS and show the latest version

3. Click the download button to get the `.pkg` installer

> Always download from python.org to ensure you get the official, secure version.

Verify installation

Open a **new** Terminal window (important!) and check the version:

```
python3 --version
```

You should see the version you just installed. The exact number depends on when you download.

Test Python

Let's make sure Python works properly:

1. In Terminal, type:

```
python 3
```

2. You'll see something like:

```
Python 3.13.5 (v3.13.5:0fa1754080, Jul 29 2025, 09:45:56) [Clang 15.0.0] on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

3. Try your first Python command:

```
print("Hello from Python!")
```

4. Press `Enter` u should see:

```
Hello from Python!
```

5. To exit Python:
   Type `exit()` and press Enter
   Or press `Ctrl + D`

Troubleshooting

'python3' command not found

The Terminal might not see the new installation yet.

Solution 1: Close Terminal completely and open a new window.

Solution 2: Check if Python is installed but not in PATH:

```
ls /Library/Frameworks/Python.framework/Versions/
```

Solution 3: Add Python to your PATH manually:

```
echo 'export
PATH="/Library/Frameworks/Python.framework/Versions/Current/bin:
$PATH"' >> ~/.zshrc
source ~/.zshrc
```

'python' command doesn't work

This is normal on macOS. You have three options:

Option 1: Always use `python3` (recommended)

Option 2: Create an alias:

```
echo 'alias python=python3' >> ~/.zshrc
echo 'alias pip=pip3' >> ~/.zshrc
source ~/.zshrc
```

Option 3: Check if the installer created a `python` link:

```
which python
```

SSL/Certificate errors

You forgot to install certificates. Fix it by

Running from Terminal:

```
/Applications/Python\ 3.*/Install\ Certificates.command
```

Or manually installing certificates:

```
pip3 install --upgrade certifi
```

Multiple Python versions

macOS can have multiple Python versions. To manage them:

See all installed versions:

```
ls -la /usr/local/bin/python*
```

Use a specific version:

```
python3.13 --version
python3.12 --version
```

Set a default (example for 3.13):

```
ln -s -f /usr/local/bin/python3.13 /usr/local/bin/python3
```

Alternative: Homebrew installation

If

Code editor

Install VS Code

Download and install Visual Studio Code

Download VS Code

Visual Studio Code is available for all operating systems. The installation is quick and straightforward.

Download & Install

1. Go to code.visualstudio.com
2. Click the big download button (it detects Windows automatically)
3. Run the downloaded installer
4. **Important:** Check these options during installation:

- ✓ Add "Open with Code" action to Windows Explorer file context menu
- ✓ Add "Open with Code" action to Windows Explorer directory context menu
- ✓ Register Code as an editor for supported file types
- ✓ Add to PATH

5. Click "Next" and "Install"
6. Click "Finish" when done

Verify installation

Once installed, let's make sure VS Code is working:

1. Open VS Code:
   - **Windows:** Search "Visual Studio Code" in Start menu
   - **macOS:** Find it in Applications or press `Cmd + Space`
   - **Linux:** Type `code` in terminal or find it in your applications menu
2. You should see the Welcome tab
3. The interface should look clean and modern

Install Python extension

VS Code needs the **Python extension** to work properly with Python files:

1. Click the Extensions icon in the left sidebar (it looks like 4 squares)
2. Search for “Python”
3. Find the one by **Microsoft** (it has millions of downloads)
4. Click “Install”
5. Wait for installation to complete

The Python extension adds syntax highlighting, code completion, debugging, and more. It’s essential for Python development.

Configure Python execution

After installing the Python extension, enable this important setting:

1. Open Settings () `Ctrl/Cmd + ,`
2. Search for “Python Terminal Execute In File Dir”
3. Check the box to enable it

What this does: When you run a Python file, VS Code will execute it from the file’s directory instead of your workspace root.

Why I recommend it:

This prevents common path-related errors. For example, if your script reads a file with `open('data.csv')`, it will look for the file in the same folder as your script, which is usually what you want.
Without this setting, it would look in your project root instead, causing “file not found” errors.

Additional recommended extensions

While VS Code works great with just the Python extension, here are a few more I recommend:

Pylance

- Search for “Pylance” by Microsoft
- Provides even better code completion and error detection
- Works alongside the Python extension

Jupyter

- Search for “Jupyter” by Microsoft
- Enables interactive Python mode (we’ll use this later)
- Essential for data science and AI work

Customize appearance (optional)

I've been using these settings for years - they're easy on the eyes:

Theme

Install "Atom One Dark Theme"

Press `Ctrl/Cmd + T`, then `Ctrl/Cmd + K` to select it

Icons

Install “Material Icon Theme” by Philipp Kief

Makes file types easier to recognize

Tree indentation

Open Settings () `Ctrl/Cmd + ,`

Search for “tree indent”

Change from 8 to 20 for clearer folder structure

`Ctrl/Cmd + Shift + P`

View keyboard shortcuts: Open Command Palette () and search “keyboard shortcuts”. You can search for any command and change its shortcut by clicking the pencil icon.

Check Python detection

After setting up VS Code:

1. Press `Ctrl + Shift + P` (Windows/Linux) or `Cmd + Shift + P` (macOS)
2. Type “Python: Select Interpreter”
3. You should see your Python installation listed
4. If not, we’ll fix this in the next section